GIDFINE — CONVEX FOUNDATION VERIFICATION AUDIT

Reference: Gidfine_Dev_Guide.md (v1.1) | **Target Branch:** development | **Audit Date:** October 19, 2026 | **Scope:** Pre-Implementation Verification

Audit Objective: Confirm the exact status of the Convex foundation changes on the development branch prior to implementing new features. Verifies git and commit status, configuration consistency, TypeScript and build compilation, test pass rates, security boundaries, and live deployment posture.

1. Git and Remote Tracking Status

Repository Check	Observed Value / State	Compliance Status
Active Branch	development (tracked to origin/development)	**PASS (Correct Branch)**
Latest Commit	7059684 docs: archive VUE_MIGRATION_PLAN.md into migration directory	**PASS (Synchronized)**
Main Branch Isolation	main remains untouched at f7ed8d1	**PASS (Zero Main Touches)**
Commit / Push Status	All foundation files reside as uncommitted changes in the working tree. **Zero commits or pushes made.**	**PASS (Constrained)**

2. Files Inventory & Modification Status

File Path	Git State	Architectural Purpose & Integrity
frontend/package.json	Modified	Added "convex": "^1.47.0" and scripts convex:dev, convex:codegen
frontend/src/main.tsx	Modified	Wrapped root application with <ConvexClientProvider>
migration/README.md	Modified	Updated migration index to register Phase 8 and Phase 9 audit reports
frontend/.env.example	Untracked	Documents VITE_CONVEX_URL and VITE_API_URL (no secrets)
frontend/src/lib/convex.tsx	Untracked	Provider wrapper with graceful pass-through when unprovisioned
frontend/convex/schema.ts	Untracked	Defines agencies, users, and immutable auditLogs
frontend/convex/auth.config.ts	Untracked	Configures JWT provider referencing environment domain
frontend/convex/lib/auth.ts	Untracked	Server authorization helpers (getVerifiedUser, requireRole)
frontend/convex/health.ts	Untracked	Connectivity probe returning system status, timestamp, and version
frontend/convex/tsconfig.json	Untracked	TypeScript configuration isolating backend function compilation
frontend/convex/_generated/	Untracked	Type definitions (dataModel.d.ts, server.d.ts, api.d.ts)

3. Verification Checks Run & Actual Results

Verification Command	Scope & Execution Details	Observed Result
npx tsc --project convex/tsconfig.json	Compiles Convex backend functions, schema, and generated types	**PASSED (0 errors)**
npm run build (tsc -b && vite build)	Full React 19 production build (1,845 modules transformed in 10.88s)	**PASSED (0 errors)**
pytest tests/ -q	Full FastAPI regression test suite (comparison, admin, mock APIs)	**6/6 PASSED (0 regressions)**
Schema & Auth Consistency	Cross-check role unions, tokenIdentifier indexing, and table models	**100% CONSISTENT**

4. Security Analysis & Incomplete Setup Disclosures

- **Live Convex Deployment Unconfigured:** There is currently **no live Convex deployment configured**. No .env.local exists with credentials. The frontend runs exclusively through the graceful fallback in ConvexClientProvider.

- **Auth Issuer Domain Placeholder:** auth.config.ts uses CONVEX_AUTH_ISSUER_DOMAIN falling back to https://auth.gidfine.com. Real tokens cannot be verified until Clerk or Auth0 is provisioned.
- **Generated File Regeneration Required:** The convex/_generated/ files were manually created for local offline type-safety. When npx convex dev is executed, the CLI will regenerate these files against the cloud deployment schema.
- **Uncommitted Working Tree:** All 13 foundation files remain unstaged and uncommitted. Any branch switch or pull without stashing risks losing this foundation.

5. Recommended Next Implementation Milestone

Milestone 1 — Cloud Provisioning: Execute npx convex dev to link the live Convex Cloud deployment and generate verified _generated/ types.

Milestone 2 — Tenancy & Identity Sign-Off: Decide whether multi-agency is required for initial prototype, and configure Clerk/Auth0 issuer.

Milestone 3 — Single-Loan Transaction Slice: Implement the complete 1-Borrower, 1-Agency, 1-Lender transaction lifecycle with integer pesewas ledger entries and server-side field masking.